

Mathematics for Data Science – True/False Review

For each statement, explain whether it is **true** or **false**.

Lecture 1: Vector spaces and linear maps

1. A subset S of a vector space V is a subspace if and only if it contains the zero vector and is closed under addition and scalar multiplication.
2. Every affine line in $\mathbb{R}^n$ is a vector subspace of $\mathbb{R}^n$.
3. If a list of vectors spans a vector space, then the list is necessarily linearly independent.
4. A linear map $L : U \rightarrow V$ is injective if and only if $\ker(L) = \{0_U\}$.

Lecture 2: Matrices and linear systems

1. If $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$, then Ax is defined and belongs to $\mathbb{R}^m$.
2. The sum $A + B$ is defined for any two real matrices, even when their dimensions are different.
3. For $A \in \mathbb{R}^{m \times n}$, the rank–nullity theorem is

$$n = \text{rank}(A) + \dim(\ker(A)).$$

4. The map $x \mapsto Wx + b$ is linear for every matrix W and every vector b .

Lecture 3: Determinants

1. The determinant is defined for every rectangular matrix.
2. For square matrices A and B of the same size, $\det(AB) = \det(A)\det(B)$.
3. Exchanging two rows of a matrix leaves its determinant unchanged.
4. A square matrix A is invertible if and only if $\det(A) \neq 0$.

Lecture 4: Eigenvalues and diagonalization

1. An eigenvector must be nonzero and satisfy $Av = \lambda v$ for some scalar λ .
2. The zero vector is an eigenvector for every eigenvalue of a linear map.
3. If an endomorphism has n pairwise-distinct eigenvalues on an n -dimensional space, then it is diagonalizable.
4. For a finite-dimensional endomorphism $f : E \mapsto E$, the characteristic polynomial has degree $\dim(E) - 1$.

Lecture 5: Normed and inner product spaces

1. An inner product on a real vector space is positive-definite: $\langle x, x \rangle \geq 0$, with equality if and only if $x = 0$.
2. A metric space must also be a vector space.
3. A norm satisfies $\|\alpha x\| = |\alpha| \|x\|$ for every real scalar α .
4. In a finite-dimensional vector space, any two norms are equivalent: there are constants $c, C > 0$ such that

$$c\|x\|_A \leq \|x\|_B \leq C\|x\|_A$$

for every vector x .